When graded PrudentBots cooperate with graded FairBots

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1 Introduction

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. A *modal agent* [1] decides by asking what Peano arithmetic (PA) can prove about the opponent, written with the provability box $\Box$ of the logic GL. Here p means "the opponent cooperates with me" and q means "the opponent cooperates with DefectBot", the agent that always defects. FairBot, $\Box p$, cooperates exactly when it can prove that the opponent cooperates with it. PrudentBot adds a safety test: it also requires a proof, in the stronger system PA+1, that the opponent would not cooperate with DefectBot, so it does not reward an opponent that cooperates with everyone.

We study two graded families. The graded FairBot $F_k = \Box^k p$ $(k \geq 1)$ asks for k nested layers of provability; F_1 is FairBot. The graded PrudentBot $P_j = \Box p \wedge \Box(\neg\Box^j \bot \to \neg q)$ $(j \geq 0)$ runs its safety test in PA+j, since $\neg\Box^j \bot$ expresses the consistency of PA+$(j-1)$; P_1 is PrudentBot.

Result

P_j and F_k cooperate with each other exactly when $k \leq j$; when $k > j$, both defect. In words: a PrudentBot that runs its safety test in a stronger system can afford to cooperate with FairBots that demand more nested proof, and the exchange rate is one for one. Every P_j still defects against CooperateBot, so the extra cooperation costs nothing in safety.

Why it matters

Barasz et al. remark that PrudentBot's safety test may use a stronger system than PA+1 and that this widens the set of agents it cooperates with, and online discussion of the paper looked at specific pairs of such variants. Here the relation is pinned down exactly for the two graded families, for all indices: the boundary of cooperation is the single inequality $k \leq j$.

Proved vs checked

Proved for all $j \geq 0$ and $k \geq 1$ in Lean 4, in the height semantics of closed GL fixed points (on the chain of heights, both agents cooperate at every height when $k \leq j$; when $k > j$ both defect above height $j+k$). The link to provability in PA is Solovay's theorem [17] (see also [8]), cited, not formalised, and the reading in the standard model assumes PA is sound.

Novelty statement

This result extends # Prior-art answer for R4 (K8: P_j / F_k threshold law)
Verdict: partial prior art.

- Barasz et al. 2014, remark after Theorem 3.2: PrudentBot's DefectBot check may use PA+n for larger n, which widens the set of agents it cooperates with without losing safety. That is the idea behind graded PrudentBots, stated informally.
- LessWrong comments on the Barasz et al. post (orthonormal, cousin_it) discuss indexed variants "Prudent(m,n)Bot" with specific pairs, e.g. Prudent(0,100)Bot cooperating with Prudent(0,1)Bot and Prudent(99,1000)Bot, and cousin_it noting that a PrudentBot using PA and PA+1 does not cooperate with one using PA+1 and PA+2.
- Not found: a general threshold law for P_j against graded FairBots F_k (cooperate iff k <= j, both defect above j+k), or a proof of it.
- Suggested wording: the graded family was discussed informally (Barasz et al. remark; LessWrong comments); the general law and its Lean proof were not found.

Sources searched (owner, web search, 2026-10-17)

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" (Alignment Forum).

- Mennen, “Modal Combat for games other than the prisoner’s dilemma” (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

Not searched: Critch’s parametric bounded Löb paper, theses, the contents of the Haskell agents folder, non-English sources. A “none” verdict means “not found in this search”, nothing more. Our prior-art search (2026-10-17) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner’s Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner’s Dilemma via Löb’s Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post “Robust Cooperation in the Prisoner’s Dilemma” and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
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If you know of related work we missed, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the proof and check files are included so anyone can re-run them.

2 Claims

Claim K8

For all j>=0, k>=1 (height-chain semantics of closed GL fixed points): P_j=[]p&[](~[]^jF->~q) and F_k=[]^k p cooperate at every height iff k<=j; if j<k both defect above height j+k. Lean-proved; P_j defects vs CB.

Status: derived.

Verified.

- `checks/K8-c046_pjfk_lean.py`: embedded Lean 4 proof of pj_fk and pj_cb compiles with no sorryAx; flipped-conclusion variant is rejected by Lean; explicit height table for j<=8,k<=8 matches (coop everywhere iff k<=j, else both defect above j+k with exact threshold) — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Closed (letter-free) GL fixed points are decided by height on finite linear frames (Boolos, Logic of Provability, ch. 7), so box phi at height h means phi at all g<h; the Lean theorem is stated in exactly this semantics for arbitrary solutions.

Prior art. Partial prior art.

Provenance. Run `20261017-1126`, claims K8, run-folder commit `662166ecff0651902cc5fcb04bf747c62d79` published 2026-10-18.

3 How to verify

Each finding has its check scripts in `claims/graded-prudent-fair/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[3] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

[4] Christoph Benzmüller. Gödel's and scott's variants of the ontological argument in lean 4 and tptp thf. arXiv:2609.26806, 2026.

[5] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

[6] Christoph Benzmüller and Dana S. Scott. Notes on gödel's and scott's variants of the ontological argument. *Monatshefte für Mathematik*, 2025.

[7] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[8] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[9] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[10] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[11] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[12] Gram_Stone. Modal chicken. LessWrong, 2015.

[13] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[14] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[15] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[16] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[17] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.